Designing Scaffolding Cards to Facilitate LLM-Based Socratic Instruction: An Exploratory Study of Response Strategies to Support Learning

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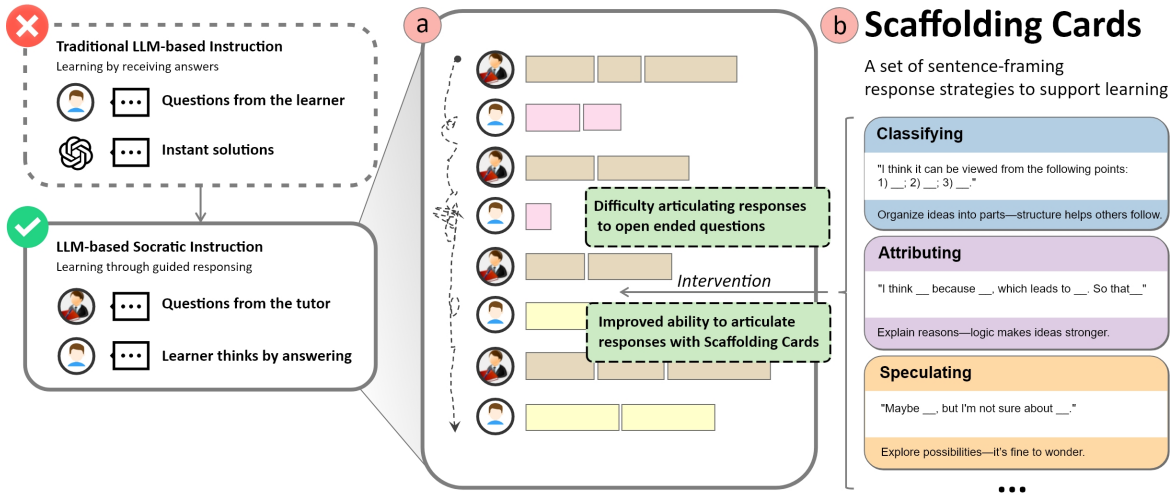


Figure 1: Overview of designing Scaffolding Cards to facilitate LLM-based Socratic instruction. a) The learners' response-construction challenges evidenced in the evolution from traditional tutor-driven interaction to LLM-based dialogue; b) Scaffolding Cards as a set of sentence-framing response strategies to support clearer expression and critical learning.

Abstract

The overreliance on large language models (LLMs)-generated answers poses risks to the development of learners' critical thinking. Socratic instruction, which follows "tutor asks, student answers" approach, could mitigate overreliance by engaging learners with LLM-generated questions rather than passively seeking answers from LLMs. However, learners without effective response strategies often produce superficial answers and therefore undermine Socratic instruction. To bridge the gap, we first conducted a formative study (N=20) to analyze learners' dialogue logs and interviews, deriving 18 Scaffolding Cards as response strategies to guide learners in framing their answers. A subsequent mixed-methods study (N=34) demonstrated that Scaffolding Cards improved critical thinking, optimized cognitive load allocation, and increased learning satisfaction compared to that without scaffolds. Our work reconfigures scaffolding by incorporating state-aware, agency-preserving, and

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function-transparent support. We further provide actionable implications for designing responsive and personalized scaffolding to facilitate learner-LLM interaction, introducing innovative perspectives for reclaiming learner agency in LLM-driven education.

CCS Concepts

• **Applied computing** → Education; Interactive learning environments.

Keywords

LLMs, Socratic instruction, learner-LLM interaction, scaffolding

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1 Introduction

Large Language Models (LLMs), as advanced artificial intelligence (AI) systems trained on vast text, could comprehend and generate human-like texts [61] and have been increasingly integrated into dialogue-based Intelligent Tutoring Systems (ITSs) [31, 67]. Within the dialogue-based ITSs, **LLMs serve as intelligent tutors** that automatically provide adaptive guidance and high-quality answers during the learner-LLM interaction [18, 76]. Therefore, **LLMs typically lead the interaction with learners** by delivering direct instruction, offering conceptual explanations, and answering questions through natural dialogue [29, 63]. While effective for knowledge dissemination, this increasingly used LLM-based ITS often prioritizes instant answers, which may reduce opportunities for learner reflection [27]. Acting as directive tutors, LLMs may lead to negative impacts on the development of learners' critical thinking, such as overreliance on the system [9, 49], illusions of understanding [56], and hindrance of skill acquisition [53].

The Socratic method is rooted in classical philosophy, following a "tutor asks, student answers" pedagogical approach that fosters critical thinking through inquiry-driven and reflective dialogue between students and instructors. With probing questions to challenge assumptions and promote deep understanding [65, 74], the Socratic method reframes the role of LLMs from answer-givers to question-askers, adopting thought-provoking prompts to elicit learner responses [21, 27]. The incorporation of the Socratic method offers a promising way to limit excessive intervention by LLMs, promoting critical thinking through open-ended questions rather than direct instructions [43, 69]. Furthermore, early studies suggest the potential of LLM-based Socratic instruction in fostering greater engagement [8] and deepening understanding of complex topics by encouraging independent thoughts [92].

Despite the abovementioned benefits, the effectiveness of LLM-based Socratic instruction in practice is contingent upon learners' active and substantive responses [36, 70]. Many prior studies have focused on the LLM-based tutor's question design [22, 92], adaptation [35, 38, 97], and sequencing [75, 97], while giving relatively little attention to how learners formulate responses. However, formulating meaningful responses to open-ended questions requires considerable effort and self-motivation [11, 16, 91]. Without mature response strategies, most learners tend to produce superficial or unoriginal answers when faced with challenging questions [39], limiting the articulation of prior knowledge and constraining subsequent learning (see Figure 1.a). The passive response weakens the dialogic learner-LLM interaction, ultimately undermining the pedagogical effectiveness of the Socratic method [8, 36].

To address the abovementioned gap, we aim to develop structured response strategies that inspire learners to effectively offer thoughtful responses, provide original statements, and synthesize existing content. We conducted a two-stage exploratory study using the Socratic Playground for Learning (SPL), a leading-edge ITS that implements LLM-based Socratic instruction [97], to address the following research questions (RQs):

- **RQ1:** How can response strategies be structured to facilitate the learner-LLM interaction in Socratic instruction?
- **RQ2:** How do response strategies impact learning outcomes in LLM-based Socratic instruction?
- **RQ3:** How do learners perceive structured response strategies in LLM-based Socratic instruction?

In the first stage, we conducted a formative study in which learners interacted with the SPL system. We identified response strategies from the dialogue logs and derived design considerations from the interview data. These insights together guided the development of **Scaffolding Cards**, a set of 18 sentence-framing response strategies that support response formulation to LLMs. The cards are organized into three clusters, including *Structuring Expression* for organizing ideas, *Constructing Reasoning* for building arguments, and *Activating Thinking* for fostering reflection and ideation (**RQ1**). Each card provides a strategy label, an editable sentence frame, and a brief description with motivational cues (see Figure 1.b), offering syntactic scaffolding while retaining agency of learners.

In the second stage, we conducted a mixed-methods study to evaluate the pedagogical effects of Scaffolding Cards compared to a control condition without scaffolds, combining dialogue logs, questionnaires, and interviews. Regarding **RQ2**, our findings showed that structured response strategies enhanced learning outcomes by deepening critical thinking and supporting personalized strategy use. These strategies enabled learners to broaden analysis, strengthen reasoning, and focus on significance, while also allowing them to tailor their choices to the dialogue context, knowledge proficiency, and personal style. Regarding **RQ3**, our findings indicated that learners perceived the strategies positively regarding the optimized cognitive load allocation and higher learning satisfaction, with learners reporting greater autonomy, clearer expression, and more confidence during dialogue. The Scaffolding Cards helped learners facilitate response initiation, focus on content, and encourage deeper engagement, shifting their efforts from wording to meaning. However, the benefits were limited when learners already had clear ideas or when the sentence frames did not match their usual expression.

This study makes the following contributions to the learning sciences community from an interaction perspective: 1) The research presents Scaffolding Cards as a novel design exemplar for

LLM-based Socratic instruction, demonstrating a set of **structured response strategies** identified in the first stage of the research, providing a practical and ready-to-use intervention to address learners' difficulties in generating substantive answers; 2) Through the mixed-method investigation using the Scaffolding Cards as a research probe, we uncover **empirical insights** from the second stage of our study, showing how scaffolding improves critical thinking, redistributes cognitive load, enhances learning satisfaction, and enables personalized strategy use in LLM-based Socratic instruction; 3) Anchored in reconfiguring scaffolding as state-aware, agency-preserving, and function-transparent support, we highlight **design implications** to facilitate learner-LLM interaction for future LLM-based Socratic instruction across four directions (i.e., adaptability, intuition, motivation, reflection). The designed scaffolding also serves as a research lens to examine how learners progress across cognitive levels and disciplinary contexts through LLM-based Socratic instruction, which offers a transformative perspective to reclaim learner agency from LLM-driven education.

2 Related Work

2.1 Opportunities and Risks of LLMs in Dialogue-based ITSs

LLMs, such as OpenAI's GPT-4 [64], are built on transformer architectures and excel in a variety of natural language processing (NLP) tasks [58], including text summarization [96], translation [95], and dialogue [87]. Recent advancements have expanded their application in education [42, 71, 93], where they demonstrate significant potential for tutoring [29, 63], automated feedback to learners [18], and personalized learning support [68]. As LLMs advance in reasoning and conversational capabilities, their integration into ITSs is increasingly enabling dynamic and engaging educational experiences [78].

Traditionally, dialogue-based ITSs have relied on rule-based algorithms to model learners' proficiency using past interactions with the system [1, 62], providing tailored feedback and dynamically adapting instructional content [50, 60]. However, such systems typically depend on pre-annotated resources and structured dialogue flows, which hinder the efficacy of adaptive tutoring in complex and unstructured learning processes [98]. The incorporation of LLMs into ITSs offers promising improvements by analyzing unstructured context and generating more human-like instructional content [51], which enhances scalability, reduces development costs, and increases learner engagement through diverse, context-sensitive feedback [4, 78].

However, certain risks are associated with the proactive tutor role of LLMs, which dominate interactions with learners in educational settings [57]. First, the overreliance on LLM-driven ITSs may foster cognitive laziness [9, 49], where learners accept AI-generated responses without critical reflection [24]. Additionally, learners may develop illusions of understanding, falsely believing they have mastered content [56]. In some cases, excessive dependence on such systems could impede the development of essential skills, potentially undermining the learning process [53]. These issues point to a breakdown in productive learner-LLM interaction, where learners become passive recipients rather than active partners in the dialogue.

2.2 The Neglect of Response Strategies in LLM-based Socratic Instruction

The Socratic method shifts the focus from obtaining instant solutions to fostering reflective dialogue that sustains deeper engagement [65, 74]. It has long been applied across diverse disciplines [45, 47, 89], including medicine [36, 66], law [77], psychology [17, 84], and business [10]. With the rise of LLMs, this method now holds promising opportunities for scale, as LLMs can serve as instructors facilitating Socratic-style dialogues [21, 39, 92]. A growing body of work has explored LLM-based Socratic instruction that can effectively replicate the inquiry-driven, dialogic nature of the Socratic method [38, 92, 97]. Subsequent studies have examined the distinctions of learner-LLM interaction between LLM-based Socratic tutors and conventional LLM chatbots, highlighting how the former foster critical thinking [21, 27] and learning engagement [8]. In addition, researchers have begun exploring domain-specific applications of LLM-based Socratic instruction, including its use in cognitive behavioral therapy [35, 41], in mathematics education to support creative problem-solving [22, 75], and in providing personalized emotional support for older adults [25].

Despite these advances, **most current studies emphasize the LLM-based tutor's questioning strategies, but relatively little attention has been given to learners' response strategies**. To address this gap, we outline three key theoretical perspectives that provide a foundation for understanding challenges in current practices. First, response quality is an essential component of learner-LLM interaction [86] and a critical factor in reflecting the effectiveness of fostering critical thinking during the Socratic method [36, 70]. Learners participating in LLM-based Socratic instruction often face open-ended questions, which require them to deepen their knowledge through self-guided reasoning [11, 16]. Consequently, learners may be more likely to skip questions or provide low-quality, even irrelevant, responses [6], which undermines the potential for critical thinking gains. Second, cognitive load is a key consideration for understanding the mental demands when learners formulate their responses. Sweller's framework [80] distinguishes Intrinsic Load arising from inherent task complexity, Extraneous Load generated by unnecessary processing demands, and Germane Load associated with meaning-making effort. In the context of LLM-based Socratic instruction, learners must both understand the open-ended questioning and articulate responses independently. This step-by-step process demands additional effort to construct meaningful answers, frequently resulting in increased cognitive load [91]. Third, learning satisfaction is tied to the perceived alignment between learning demands and the supportiveness of the environment [101], which collectively influences engagement and persistence. The absence of a human instructor acts as a double-edged sword for how learners formulate their responses: while it removes the pressure and anxiety often associated with classroom dynamics [2, 23], it may also reduce learners' sense of urgency and responsibility. Furthermore, the increased learner autonomy in selecting learning content often clashes with limited prior knowledge [75], making it challenging to generate satisfying responses.

Therefore, this paper redirects attention to the learner's role in **constructing responses** during Socratic dialogue and examines **response strategies** that support the formulation of thoughtful

answers, aiming to foster critical thinking, productive cognitive load, and learner satisfaction in LLM-based Socratic instruction. The three aims align with the three abovementioned theoretical perspectives, which collectively provide the grounding for the hypotheses developed in section 4.3.4.

2.3 Scaffolding as an Approach for Response Construction Challenges

Scaffolding is first designed to provide support for learners, but only enough to allow the learner to complete the task alone [7, 33]. It is closely aligned with Vygotsky's [85] sociocultural theory, particularly the Zone of Proximal Development, which emphasizes the range between what a learner can do independently and what they can achieve with guidance. In education, scaffolding is recognized as a key instructional strategy that supports learners by simplifying complex tasks in ways that ease cognitive demands [3] while fostering autonomy and meaningful engagement [40]. Its applications span a variety of educational contexts, such as improving English learners' speaking performance [73], supporting gamified lesson design through cognitive tools [15], accommodating diverse needs in learner-centered physical education [26], fostering mathematical thinking through inquiry-based learning [13], and enhancing engagement in authentic scientific inquiry [83].

As technology extends learning beyond the classroom into intelligent systems, scaffolding is no longer limited to face-to-face teacher-student interactions [79]. In AI-powered learning environments, scaffolding increasingly supports learner-LLM interaction, helping learners engage more productively with intelligent systems through dialogic collaboration. In the context of this study, we explore how scaffolding can be incorporated into an ITS that employs LLM-based Socratic instruction. This form of support, termed as scaffolding for dialogue [44], focuses on the ways in which the LLM-based Socratic tutor facilitates deeper reasoning through open-ended questioning. Such questions "open a dialogic space" [88], encouraging learners to consider diverse possibilities and engage in higher-order thinking. The primary goal of our response strategy design is to widen the dialogic space, for instance, by prompting learners to introduce new perspectives [44].

Scaffolding Card is a particularly promising form of scaffolding in dialogic learning environments. Prior research has shown the effectiveness of card-based tools in structuring inquiry and collaboration. For instance, Moore [59] addressed students' discomfort with challenging peer ideas by introducing sentence frames and back-pocket question cards, which improved both self-efficacy and students' capacity for critical argumentation. Toh et al. [82] used Scaffolding Cards as cognitive supports during early problem-solving stages. Littlefield et al. [52] introduced We! Connect Cards to enhance interpersonal connections in both educational and workplace settings. Yeoman [94] developed ACAD cards as a design scaffold, leveraging revised Bloom's taxonomy [5] to guide learning design across various phases of the design-thinking process. Similarly, Tahir et al. [81] explored how design cards can support creativity and collaboration in early stages of game-based learning (GBL) design workshops.

Inspired by this body of work, this study adopts **a Scaffolding Card approach** to support learner responses in LLM-based Socratic instruction. This approach further allows us to examine how scaffolding can be situated within learner-LLM interaction in dialogic learning processes.

3 STAGE 1: Formative Study

To address **RQ1**, we conducted a formative study to investigate how learners interacted with the SPL system and what forms of support were needed to facilitate meaningful engagement. This investigation led to a set of response strategies and design considerations that informed the design of Scaffolding Cards in LLM-based Socratic instruction.

3.1 Participants and Procedure

3.1.1 Demography of Participants. We recruited 20 participants (P1-P20) through the university network. To ensure sample diversity and gain broader insights into response strategies, participants were selected from various academic disciplines and stages, with a mean age of 23.45 ($SD = 2.46$), ranging from 21 to 30 years, comprising 8 males and 12 females. All participants provided informed consent and completed a demographic questionnaire, including details such as name, gender, major, academic stage, and experience using LLMs. To ensure valuable data collection, participants were required to have at least one year of prior experience using LLM chatbots, such as ChatGPT, for personal learning. The detailed participant demography is presented in Table 1.

3.1.2 Dialogue on the Socratic Playground for Learning. Building on prior work, we used the previously developed Socratic Playground for Learning (SPL) [97], a dialogue-based ITS powered by GPT-4 that operationalizes the Socratic method to cultivate critical thinking through multi-turn interaction. SPL uses prompt-engineered templates to automatically construct personalized learning scenarios from text or tree-structured inputs, then conducts iterative question dialogues with adaptive, context-sensitive feedback. The system emphasizes personalization, adaptability, and cross-domain coverage, supporting learning across computer science, business, engineering, psychology, nursing, mathematics, physics, and economics. These properties made SPL an appropriate system for our investigation into inquiry-based, learner-LLM dialogue.

After providing informed consent, participants received guidance on using the SPL system and completed tasks in two stages. The SPL system supports usage in two main stages: 1) **constructing learning scenarios**. The participants created learning scenarios through text-based task descriptions (see Figure 2.a). For example, P2 initiated her learning request as: *"I am working on my thesis and trying to grasp machine learning principles and applications. Any foundational insights or real-world examples would be greatly appreciated."* 2) **engaging in an interactive dialogue environmen**t. Participants engaged in a 10-round question-and-answer dialogue with the LLM-based Socratic tutor. The process began with the SPL system presenting a relevant example scenario and posing an initial question related to the learning request. After capturing the participant's response and dialogue history, the system evaluated their understanding, provided timely feedback, and posed a follow-up probing question to encourage deeper reflection (see Figure 2.b). For instance, a dialogue might begin with the tutor asking, *"What do you think is the purpose of a neural network in machine learning?"*

Table 1: Participant demography of the formative study.

Participant	Gender	Major	Academic stage	Experience using LLM
P1	Male	Biochemistry	Postdoctoral	2-3 years
P2	Female	Ergonomics	Doctoral	1-2 years
P3	Male	Vehicle Engineering	Undergraduate	1-2 years
P4	Female	Earth Sciences	Undergraduate	1-2 years
P5	Female	Urban Planning	Doctoral	2-3 years
P6	Female	Design Studies	Master	2-3 years
P7	Female	Product Design	Undergraduate	2-3 years
P8	Female	Design Studies	Master	2-3 years
P9	Female	Psychology	Undergraduate	1-2 years
P10	Female	Industrial Design	Undergraduate	1-2 years
P11	Male	Computer Science	Doctoral	2-3 years
P12	Male	Industrial Design	Master	1-2 years
P13	Female	Industrial Design	Undergraduate	1-2 years
P14	Female	Design Studies	Master	1-2 years
P15	Male	Product Design	Undergraduate	1-2 years
P16	Male	Product Design	Undergraduate	1-2 years
P17	Female	Design Studies	Doctoral	2-3 years
P18	Male	Industrial Design	Undergraduate	1-2 years
P19	Male	Architecture	Undergraduate	1-2 years
P20	Female	Industrial Design	Undergraduate	1-2 years

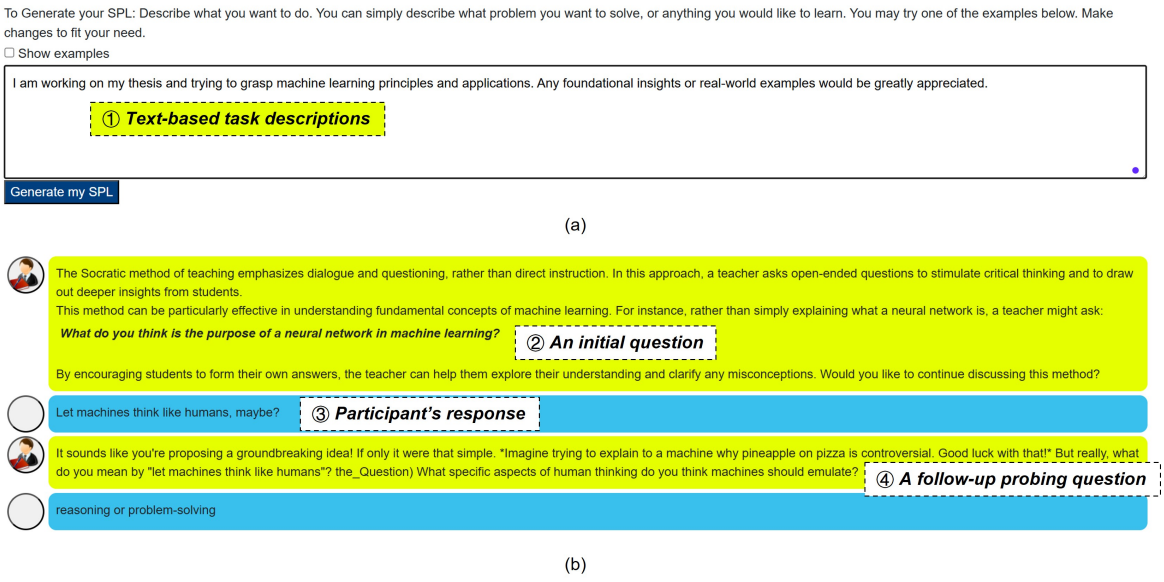


Figure 2: Workflow of the SPL system: a) Learning scenario input; b) Socratic tutoring dialogue.

to which a participant responded, *"Let machines think like humans, maybe?"* followed by the tutor probing, *"What specific aspects of human thinking do you think machines should emulate?"* In total, we collected 20 dialogue logs, comprising 200 answers from the participants.

3.1.3 Interviews. After completing their interactions with the SPL system, participants took part in a semi-structured interview aimed at understanding 1) their learning motivations and familiarity with selected learning scenarios, e.g., *"why did you choose this learning topic? how familiar are you with it?"*, 2) response challenges and confident moments during Socratic dialogues, e.g., *"what were the main challenges you encountered while responding to questions in the dialogue?"*, and 3) expected support for structuring responses, e.g., *"what specific features or functionalities would you expect from support tools (if any)?"*. Follow-up questions were used flexibly to probe specific examples and clarify responses. Each interview lasted

Table 2: Cognitive levels of Bloom's revised taxonomy with descriptions and coded examples.

Levels	Descriptions	Coded Examples
1. Remember	Retrieving relevant knowledge from long-term memory.	"This task is rhythm-based" (P15's Answer 9)
2. Understand	Determining the meaning of instructional messages, including oral, written, and graphic communication.	"For example, a virtual co-driver might help build trust in autonomous driving." (P3's Answer 4)
3. Apply	Carrying out a goal or using a procedure in a given situation.	"To understand a monocentric structure, I can look into its historical background and supporting theories." (P5's Answer 7)
4. Analyze	Breaking material into its constituent parts and detecting how the parts relate to one another and to an overall structure or purpose.	"Under high tariffs, import prices rise. Although quality and brand remain the same, consumers may shift to domestic products after weighing cost-effectiveness." (P18's Answer 10)
5. Evaluate	Making judgments based on criteria and standards.	"In my experience, data must be representative and free of extreme outliers that distort results. It should also be authentic, not tailored to fit assumptions." (P2's Answer 4)
6. Create	Putting elements together to form a novel, coherent whole, or to make an original product.	"If team and individual rewards can be separated, the highest-scoring team is rewarded, but within each team, low-scoring members face penalties." (P7's Answer 10)

approximately 20 minutes and was audio-recorded with consent for subsequent analysis. The detailed interview protocol is provided in Appendix A.1.

3.2 Data Analysis

We first conducted a deductive analysis of dialogue logs based on Bloom's revised taxonomy, and then inductively analyzed the content of learners' answers to identify potential response strategies. In parallel, an inductive thematic analysis of interview transcripts yielded complementary insights that were triangulated to refine the strategies and derive design considerations. Together, these informed the development of Scaffolding Cards.

3.2.1 Hybrid Deductive-Inductive Analysis of Dialogue Logs. We adopted Bloom's revised taxonomy [5] because it provides a well-established framework for examining the cognitive processes underlying learning. This is particularly suitable for our SPL system, where dialogues can involve and support student learning across different cognitive levels, thereby allowing us to analyze responses in terms of their position within the taxonomy. The taxonomy outlines six hierarchical cognitive levels: Remember, Understand, Apply, Analyze, Evaluate, and Create (Table 2), while answers indicating a lack of engagement or knowledge (e.g., "I don't know" or no attempt to answer) were flagged as a separate category outside the taxonomy in this study. Guided by Bloom's revised taxonomy, two researchers independently conducted a deductive analysis of the 200 answers to examine the cognitive levels during Socratic dialogues. Subsequently, they carried out an inductive thematic analysis of the categorized data to identify potential response strategies. We assessed intercoder reliability using Cohen's kappa, resulting in a value of 0.764, indicating substantial agreement [54]. The remaining discrepancies were resolved through discussion to reach consensus on final coding assignments. The finalized response strategies are detailed in section 3.3.1.

3.2.2 Inductive Analysis of Interviews. We adopted an inductive, bottom-up thematic approach [12] to analyze the interview transcripts. Meaningful data segments were first identified from the interview transcripts, and two researchers then independently coded a randomly selected 50% subset of the interview data [19], applying open coding to identify descriptive segments. After completing this round of coding, the researchers met to compare their interpretations, resolve discrepancies through discussion, and collaboratively develop an initial codebook. Using the finalized codebook, the remaining 50% of the transcripts were then independently coded by the same researchers, yielding a Cohen's kappa of 0.883, indicating almost perfect interrater reliability [54]. After completing the full coding process, all codes were reviewed and grouped into broader thematic categories. These categories not only served to refine and validate the response strategies identified from dialogue logs but also synthesized into three overarching design considerations, which are presented in section 3.3.2.

3.3 Findings

To answer **RQ1** on structuring response strategies for learner-LLM interaction, we first derived concrete strategies from the dialogue and interview analysis (3.3.1), then identified design considerations from interview insights (3.3.2), and finally integrated these into the design and use of Scaffolding Cards (3.3.3).

3.3.1 Response Strategies Derived from Analysis. Our deductive analysis of the dialogue logs showed that 177 answers were coded as valid, while 23 answers were classified as non-engaged. This finding empirically confirms the research gap identified in this study, namely that learners may face challenges in constructing responses to open-ended questions. The inductive analysis of dialogue logs, supplemented by interview data, revealed common patterns of response content across cognitive levels.

Remember: Most learners used the *simplifying* strategy to recall prior knowledge by condensing information from memory into key

Table 3: Summary of response strategies with sentence frames.

Cognitive levels	Response strategies	Sentence frames
Remember	Simplifying	"In short, I think __."
Understand	Classifying	"I think it can be viewed from the following points: 1) __; 2) __; 3) __."
	Exemplifying	"For example, __. This shows that __."
	Summarizing	"In summary, __, which indicates __."
	Clarifying	"In other words, __ refers to __."
	Analogizing	"This is like __, which helps explain __."
Apply	Implementing	"In the context of __, I think __."
	Targeting	"In order to __, I can __."
Analyze	Attributing	"I think __ because __, which leads to __. So that __"
	Differentiating	"On the one hand, __; on the other hand, __."
	Organizing	"At first I thought __, then I found __, so now I believe __."
	Countering	"However, a counter-example would be __, so __."
Evaluate	Reflecting	"My experience is __, which made me realize __."
	Critiquing	"Although __, __ is also __ because __."
Create	Speculating	"Maybe __, but I'm not sure about __."
	Hypothesizing	"If __, then __, which shows that __."
	Challenging	"Does this mean that __? If so, __."
	Questioning	"Could you tell me __?"

related phrases. For example, P4 recalled the relevant concept and articulated it as *"light-shadow relationship"* when responding to the question, *"What is the effect of the light circle on the photography?"*

Understand: Some learners demonstrated understanding by using strategies such as *classifying* to categorize their responses into numbered points or *exemplifying* to provide concrete examples in their responses. For example, P8 noted, *"I classified my thoughts into key aspects and then explained them point by point."* Others engaged in using *summarizing* to draw interim conclusions based on previous dialogues, such as a response from P14: *"As just discussed, the essence of the teaching process ultimately comes down to teacher guidance."* Additionally, some learners employed *clarifying* to restate concepts in their own words or *analogizing* to explain one unfamiliar concept through its similarity to another.

Apply: Learners typically applied knowledge to achieve specific aims in their responses, either by *implementing* to apply concepts in a given context or by *targeting* to take actions to reach a goal. For instance, when asked about a monocentric structure in urban planning, P5 applied the concept with the goal in mind, stating, *"To understand a monocentric structure, I can examine its historical background and supporting theories."*

Analyze: Learners strengthened their analytical reasoning by using strategies such as *attributing* to reason about the causes of an issue or concept, *differentiating* to compare two aspects of a concept, *organizing* to sequence the change in their thoughts, and *countering* to give counter-examples to show exceptions. These approaches were helpful for learners with weaker analytical ability. For example, P1 remarked, *"It was hard to decide which factors were most important,"* and P18 said, *"Too much information slowed me down."*

Evaluate: Learners used strategies such as *reflecting* on personal experiences to subjectively judge claims and *critiquing* to point out another factual evidence to objectively evaluate claims. For example, P4 evaluated the task by reflecting on prior experiences, saying, *"Even without a clear direction, I tried to connect my past experiences,"* while P14 described the dialogue as *"feeling like a debate."*

Create: Some learners engaged in creative thinking by using strategies such as *speculating* to envision potential outcomes they are unsure about and *hypothesizing* to construct more realistic scenarios or solutions for the test. Others advanced the dialogue by *challenging* to formulate novel assumptions or *questioning* to initiate further interactive discussion. For example, P11 explained, *"When facing repetitive questions that felt like a dead loop, I kept asking the LLM new questions to change the flow."*

As shown in Table 3, we finally synthesized and described the concrete response strategies as sentence frames inspired both by Moore [59] and by learners' own suggestions for more explicit frameworks (e.g., P5).

3.3.2 Design Considerations for Cards. Based on the inductive analysis of interview transcripts, we derived three key design considerations (DCs) for the Scaffolding Cards: 1) scaffolding with adaptation to learner states, 2) scaffolding with preservation of learner agency, and 3) scaffolding with transparent functions in LLM-based Socratic instruction.

DC1: Scaffolding with adaptation to learner states. The cards should address three typical states in which learners approach open-ended questions. 1) In familiar situations (35.0% *"fairly familiar,"* 10.0% *"very familiar"*), learners aimed to review or validate ideas, such as improving academic performance (P3, P8, P13, P18), yet often faced information overload that hindered efficient expression (P18), pointing to a need for tools that help with *Structuring Expression*. 2) In semi-familiar situations (25.0% *"somewhat familiar"*), learners pursued grounded exploration, such as advancing career skills (P16) or mastering tools (P19), but struggled when abstract questions were difficult to interpret (P1, P16), indicating a

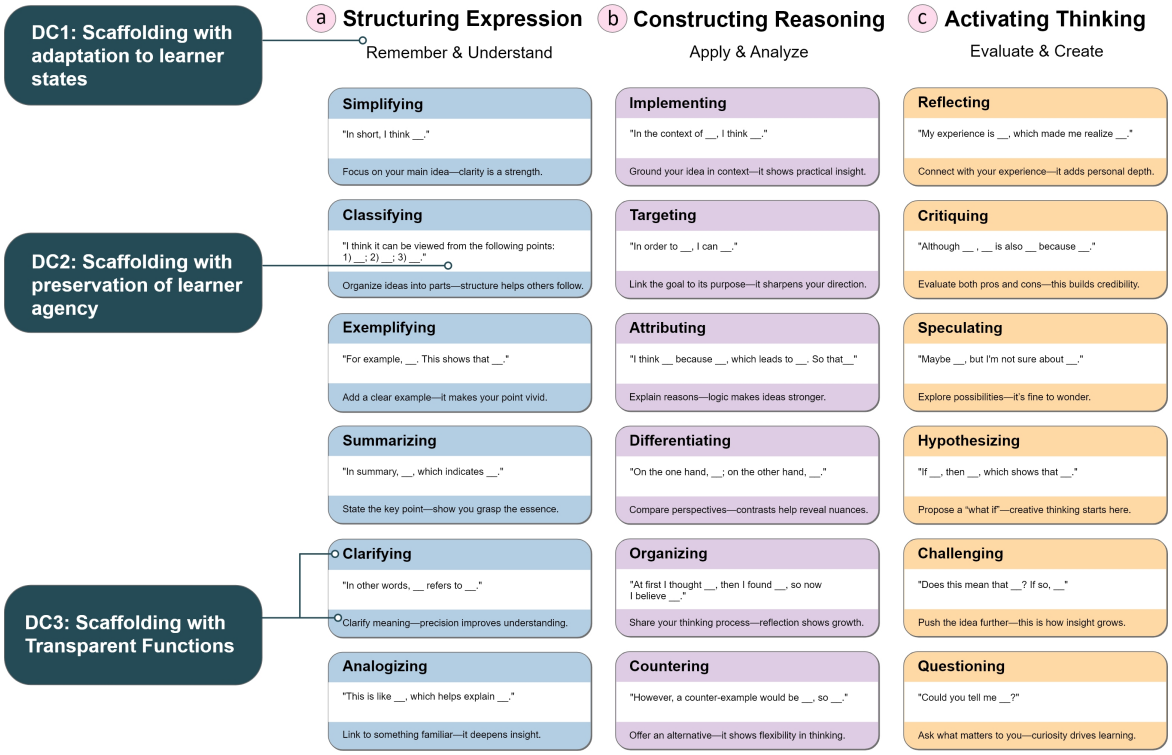


Figure 3: Overview of Scaffolding Cards organized into three clusters: a) Structuring Expression, b) Constructing Reasoning, and c) Activating Thinking.

need for *Constructing Reasoning*. 3) In unfamiliar situations (30.0% *"completely unfamiliar"*), learners were motivated by curiosity (P9, P14) or recent experiences (P11) but lacked prior knowledge (P2, P6, P9, P11), indicating a need for *Activating Thinking* strategies that stimulate imagination and support exploratory responses.

DC2: Scaffolding with preservation of learner agency. The cards should provide structured yet non-restrictive support that lowers formulation effort while retaining the agency of learners. Nineteen of 20 participants indicated that support tools would be helpful, and several asked for structured guidance such as learning frameworks (P3, P5, P11) and sentence templates or response suggestions (P8, P10, P13, P17, P18). While some participants suggested adding brief prior-knowledge cues (P6, P9, P11), a minority cautioned against overreliance, noting that excessive structure could constrain proactive thinking and reduce engagement (P2, P19). These interview insights suggest the practicality of sentence frames introduced in 3.3.1 by indicating a need for structure that does not supply content. Accordingly, the response strategies are implemented as concise, optional, and editable sentence frames that provide syntactic scaffolding while leaving semantic content fully under learners' control.

DC3: Scaffolding with transparent functions. The cards should make their scaffolding function explicit so that learners can better sustain motivation by avoiding disengagement and maintaining commitment to the learning goal. Several learners reported a gradual decline in motivation when the structure or feedback was unclear. For example, P20 noted that *"open-ended questions without any clear target made me lose patience."* Others emphasized the need for clearer articulation of the dialogue's purpose (P7, P14) and more visible indicators of prompt (P19) to stay engaged. To address these issues, each card should first provide a clear title that conveys the associated response strategy, enabling learners to quickly locate and apply it. In addition, the card text can include concise explanations of the strategy's purpose, helping learners reconnect with their intent when responding, together with brief encouraging or affirming phrases that acknowledge effort and persistence. These micro-messages make the scaffolding more transparent and help stabilize motivation during moments of uncertainty.

3.3.3 Design of Scaffolding Cards. Guided by the design considerations, we co-designed the response strategies with learners to form a set of 18 Scaffolding Cards (see Figure 3). The cards are organized into three clusters, including *Structuring Expression*, *Constructing Reasoning*, and *Activating Thinking* (DC1). Each cluster consists of six cards. Each card provides a concise sentence frame (DC2), accompanied by a strategy label and a short description with motivational cues (DC3).

- *Structuring Expression* targets familiar problems and helps learners organize initial ideas so they can begin clearly with minimal effort. It comprises strategies from **Remember** and **Understand**: *Simplifying*, *Classifying*, *Exemplifying*, *Summarizing*, *Clarifying*, and *Analogizing*.

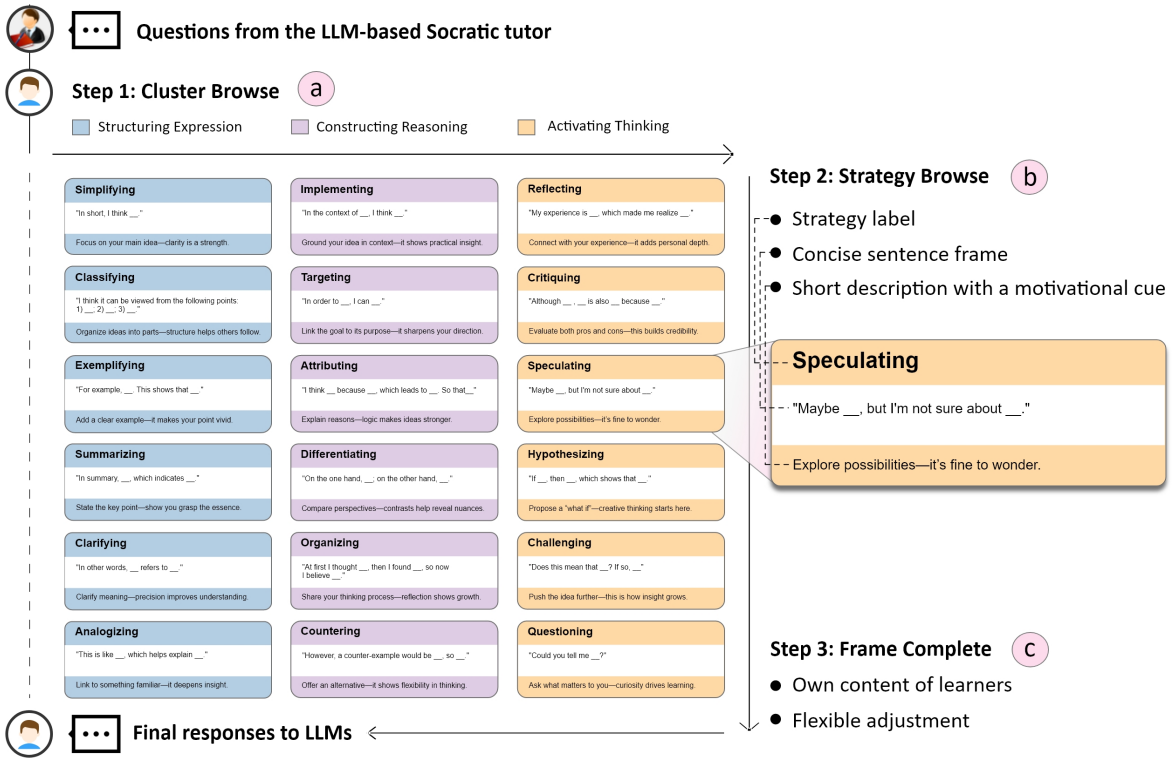


Figure 4: A recommended three-step procedure for card usage: a) Cluster Browse, b) Strategy Browse, and c) Frame Complete.

- *Constructing Reasoning* targets semi-familiar problems and helps learners apply knowledge and analyze relations to form reasoned claims. It comprises strategies from **Apply** and **Analyze**: *Implementing, Targeting, Attributing, Differentiating, Organizing,* and *Countering*.
- *Activating Thinking* targets unfamiliar problems and prompts higher-order engagement for judgment and ideation. It comprises strategies from **Evaluate** and **Create**: *Reflecting, Critiquing, Speculating, Hypothesizing, Challenging,* and *Questioning*.

Usage recommendations. Within each question-and-answer turn, learners use the cards through a **three-step** procedure, as shown in Figure 4: 1) **Cluster Browse:** Learners first self-assess their familiarity with the question and consult the cluster aligned with that state: *Structuring Expression* for familiar, *Constructing Reasoning* for semi-familiar, *Activating Thinking* for unfamiliar; 2) **Strategy Browse:** Learners skim the strategy labels within the chosen cluster to shortlist options, review sentence frames and brief descriptions, and compare them with the question and their emerging plan to select a suitable card, while remaining free to draw alternative strategies from other clusters if needed; 3) **Frame Complete:** After selecting a card, learners complete the sentence frame with their own content to produce the final response, with the flexibility to adjust or refine the sentence structure so that the answer aligns more naturally with the question.

4 STAGE 2: User Study

Leveraging Scaffolding Cards as a research probe, we conducted a mixed-method user study to evaluate the effectiveness of the response strategies developed in the formative phase, aiming to examine both the effects of the cards on their learning outcomes (**RQ2**) and learners' perception of the cards (**RQ3**).

4.1 Participants

We recruited 34 participants through the university network to take part in the user study. The required sample size was determined using G*Power for a paired-sample t-test, assuming a medium effect size (*Cohen's* $d_z = 0.5$), a statistical power of 0.8, and a significance level of 0.05. Participants were university students from a range of academic disciplines (e.g., product design, computer science, mechanical engineering, economics, etc.), with a mean age of 23.76 years ($SD = 3.09$), comprising 14 males and 20 females. None of them participated in the formative study. Prior to the study, all participants provided informed consent. They were briefed on the study's purpose, procedures, data protection protocols, and their right to withdraw at any time without penalty. In addition to completing a demographic questionnaire, participants reported their prior experience with LLMs such as ChatGPT via a 5-point Likert-scale survey assessing their proficiency in using chatbots for learning support. Results indicated a moderately high level of chatbot usage proficiency across the sample ($M = 3.79$, $SD = 0.88$), suggesting that participants were generally familiar with AI-mediated learning environments.

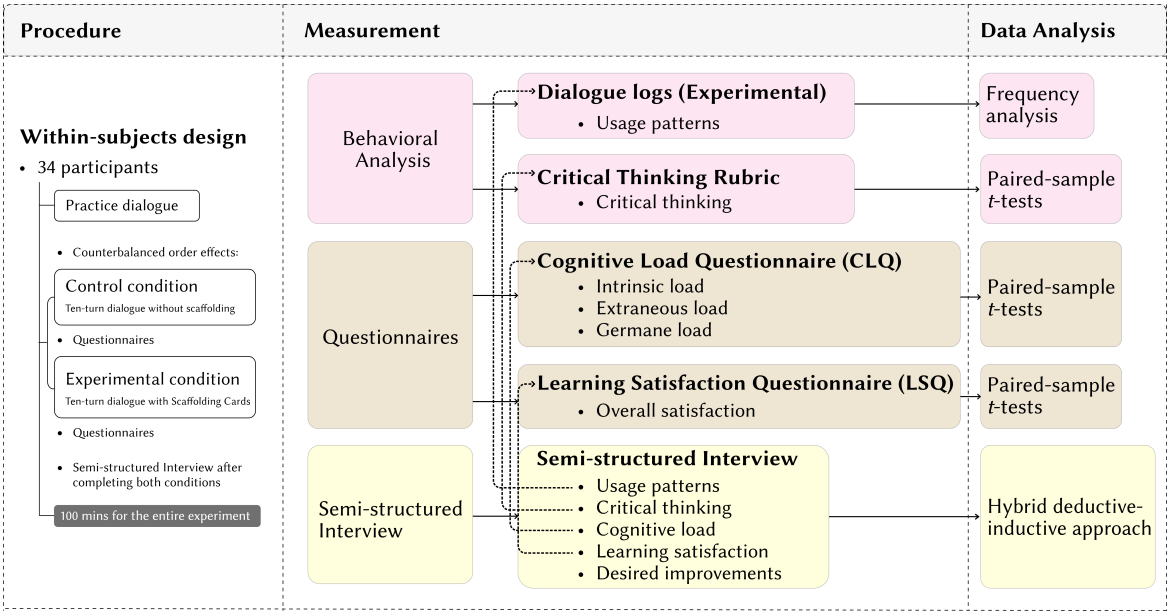


Figure 5: The framework of the user study

4.2 Procedure

The user study also employed the SPL system, enabling both learning scenario construction and Socratic dialogue with an LLM-based tutor. At the start, participants completed a five-turn practice dialogue to familiarize themselves with the SPL interface. They selected a learning topic of personal relevance and interacted with the system without any scaffolding support. This practice session was excluded from the final analysis, and participants continued with the same learning topic in the subsequent formal session. Figure 5 illustrates the overall framework of the user study.

The study followed a **within-subjects design**, in which each participant completed both a control and an experimental condition. To mitigate order effects, participants were randomly and equally assigned to one of two counterbalanced groups: one group completed the control condition first, while the other began with the experimental condition. The effectiveness of counterbalance was verified with an independent samples t-test on the two order groups (see Appendix B). No significant differences were observed across dependent variables ($p > .05$), indicating that order effects were effectively controlled. A two-minute break was provided between the two sessions to reduce fatigue and cognitive carryover. In the **control condition**, participants continued the dialogue for ten turns without access to any support tools. In the **experimental condition**, participants were introduced to a set of Scaffolding Cards offering sentence-framing strategies to aid response construction, accompanied by brief guidance on their use. They then engaged in another ten-turn dialogue, where they were encouraged to refer to the cards and use one or more sentence frames as needed, without being required to apply them rigidly. Participants browsed the entire set of Scaffolding Cards independently, without receiving any recommendations or highlighted suggestions.

The ten-turn limit was imposed to control fatigue and ensure session feasibility, with the average session duration being 38.86 minutes ($SD = 9.15$) for the experimental condition and 30.53 minutes ($SD = 7.37$) for the control condition. The time difference appears to result from the process of consulting the Scaffolding Cards, as participants needed to read, select, and integrate appropriate sentence frames into their responses, thereby extending the response construction process. To assess the potential impact of the detected time difference, linear regressions were conducted using the time difference as a predictor of each dependent variable's difference scores (see Appendix C). None of the regressions were significant ($p > .05$), indicating that the additional time spent using the cards did not systematically influence the dependent variables. In total, we collected 34 dialogue logs, comprising 680 answers from the participants. Each condition was followed by a post-task questionnaire evaluating the participant's experience. At the end of the sessions, participants completed a semi-structured interview reflecting on their experiences across both conditions and their perceptions of the Scaffolding Cards. The entire experiment took approximately 100 minutes to complete.

4.3 Data Collection and Analysis

4.3.1 Dialogue Analysis. To answer **RQ2**, we conducted a dialogue analysis focusing on learning outcomes, examined through two aspects: the frequency of strategy usage in the experimental condition and the level of critical thinking demonstrated in learner responses.

First, we conducted a frequency analysis of strategy usage patterns based on the 340 answers from dialogue logs generated under the experimental condition. For each response, we recorded which Scaffolding Card strategies were used, allowing us to quantify usage frequency across all 18 response strategies.

Second, to evaluate learners' critical thinking, response qualities of all 680 answers (340 per condition) were independently evaluated by two researchers following a critical thinking rubric adapted from Ho et al. [36], which includes nine binary-scored dimensions: clarity, accuracy, precision, relevance, depth, breadth, logic, significance, and fairness. Each dimension was scored as either 0 (skill not demonstrated) or 1 (skill demonstrated). A third researcher reviewed the response, and the final score was determined after reaching consensus to address the disagreement. The detailed rubric is provided in Appendix D. The final critical thinking score for each participant was calculated by first summing the nine binary-scored dimensions for each of their ten responses and then averaging these ten scores to obtain a participant-level mean critical thinking score. Differences between the control and experimental conditions were tested using paired-sample t-tests. The level of statistical significance for all analyses was set at 0.05.

4.3.2 Questionnaires. To answer **RQ3**, we administered two post-task questionnaires on a 7-point Likert scale to assess learners' perceived cognitive load and learning satisfaction. The **Cognitive Load Questionnaire (CLQ)** was developed based on Sweller's [80] three-factor model of cognitive load theory: 1) Intrinsic Load, reflecting the inherent complexity of the task demands; 2) Extraneous Load, arising from task-irrelevant processes; and 3) Germane Load, representing the effort directed toward meaning-making. Each dimension was measured using three items, and the average score of each dimension was used for analysis. The **Learning Satisfaction Questionnaire (LSQ)** was adapted from previous research [28, 97] on Socratic learning environments and consisted of three items, measuring overall learning satisfaction. The specific items of the two questionnaires are listed in Appendix E. Both CLQ and LSQ demonstrated acceptable reliability and construct validity (CLQ: *Cronbach's* $\alpha = 0.773$, *KMO* $= 0.804$; LSQ: *Cronbach's* $\alpha = 0.954$, *KMO* $= 0.733$). Questionnaire ratings were compared across conditions using paired-sample t-tests with a significance threshold of 0.05.

4.3.3 Semi-structured Interview. Following the experiment session, we conducted semi-structured interviews to further examine participants' experiences with the Scaffolding Cards. The interview questions were aligned with the indices of dialogue analysis and questionnaires, including 1) usage patterns of cards, 2) critical response development, 3) variations in cognitive load, and 4) perceived learning satisfaction. In addition, we asked about 5) desired improvements to gather insights into future design. The full interview protocol is provided in Appendix A.2. All interviews were conducted via audio conferencing and recorded with participant consent. Transcripts were automatically generated, then manually reviewed and corrected by two researchers.

We employed a hybrid deductive-inductive approach to analyze interview transcripts. First, we deductively structured our analysis with predefined analytic dimensions, including **usage patterns**, **critical thinking, cognitive load, learning satisfaction,** and **desired improvements**, which guided the synthesis of insights from participants. Second, we conducted the thematic analysis to inductively identify reasons behind varying perceptions. Two researchers independently performed open coding on the transcripts, compared code sets, and resolved discrepancies to produce a shared codebook. Using this codebook, they re-examined the full corpus and iteratively refined codes via constant comparison until thematic saturation. The detailed codes are summarized in Appendix F and introduced in section 4.4, with the exception of the codes for desired improvements, which are introduced separately in section 5.2 as the basis for deriving design implications.

4.3.4 Hypotheses. Drawing on prior literature on the critical thinking assessment, cognitive load theory, and learner satisfaction in Socratic instruction (see section 2.2), we developed the following hypotheses for evaluating the pedagogical effects of Scaffolding Cards. The hypotheses are aligned with the three dimensions of our comparative evaluation: critical thinking, cognitive load, and learning satisfaction, and they correspond directly to our research questions.

Hypothesis1 (**H1,** addressing RQ2): Critical thinking in the Socratic method is reflected primarily through the quality and completeness of learners' responses [36]. Building on this perspective, since the structured response strategies provided by Scaffolding Cards are designed to help participants articulate their ideas more effectively, we believe that the cards potentially enhance the overall quality of responses, thereby supporting critical thinking. Accordingly, **H1** predicts that critical thinking scores will be significantly higher in the experimental condition than in the control condition.

Hypothesis2 (**H2**, addressing RQ3): First, Intrinsic Load is theorized to depend primarily on the inherent complexity of the task materials [80]. In this context, since participants engage with identical topics across both conditions in this within-subject design, we expect that substantial differences in intrinsic complexity are unlikely. Thus, **H2a** predicts that Intrinsic Load ratings will not differ between the experimental and control conditions. Second, Extraneous Load arises from avoidable cognitive processes unrelated to the task's core demands [80]. Following this rationale, we expect that providing sentence frames may reduce the effort required to transform ideas into coherent responses, potentially decreasing extraneous processing. Therefore, **H2b** predicts that Extraneous Load ratings will be significantly lower in the experimental condition than in the control condition. Third, Germane Load reflects cognitive resources devoted to productive learning [80]. Building on this definition, since the Scaffolding Cards are designed to guide learners toward more structured reasoning, we expect that they may increase the cognitive effort invested in meaning-making. Accordingly, **H2c** predicts that Germane Load ratings will be significantly higher in the experimental condition than in the control condition.

Hypothesis3 (**H3**, addressing RQ3): Satisfaction reflects the perceived alignment between learning demands and the supportiveness of the learning environment [101]. Following this idea, since the Scaffolding Cards are likely to reduce unproductive effort and support clearer response construction, we assume that learners may perceive the experience as more helpful and engaging. Thus, **H3** predicts that overall satisfaction ratings will be significantly higher in the experimental condition than in the control condition.

4.4 Results

To answer **RQ2** on how these strategies impact learning outcomes, we report usage patterns of the Scaffolding Cards and critical thinking scores (sections 4.4.1 and 4.4.2). To answer **RQ3** on learners'

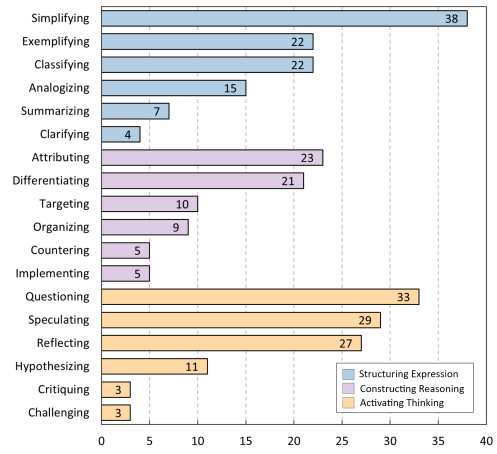


Figure 6: Frequency of strategy usage across the 18 Scaffolding Cards during the experimental condition.

perceptions of structured response strategies in LLM-based Socratic instruction, we report cognitive load and learning satisfaction results (sections 4.4.3 and 4.4.4).

4.4.1 Usage Patterns of Scaffolding Cards. In the experimental condition, participants used the cards extensively across turns. Usage spanned all three clusters: *Structuring Expression* (n = 108), *Constructing Reasoning* (n = 73), and *Activating Thinking* (n = 106). Most responses incorporated at least one card, while only one participant (P8) used none. Across strategies, the top three by frequency were *Simplifying* (n = 38), *Questioning* (n = 33), and *Speculating* (n = 29). Detailed per-strategy counts are reported in Figure 6. It is evident that most participants in the experimental group actively chose to use the cards, which aligns with the longer session duration observed in this condition.

Interview evidence revealed that usage patterns were purposeful rather than random, shaped by three factors:

- **Dialogue context calibrates immediate purpose**. Learners adapted strategies not only to the question type but also to their position within the dialogue sequence. For example, several participants (P7, P19, P27) described *Simplifying ("in short")* or *Reflecting ("my experience is...")* cards as useful starting points that helped them initiate conversation. The *Questioning* card was often used to redirect or probe the dialogue, as illustrated by P17 (*"steering the dialogue back to the focus"*) and similarly noted by P34. The *Attributing* card clarified cause-and-effect logic and was said to *"unpack deeper motivations"* (P21), a view echoed by P31. Likewise, the *Differentiating* card enabled comparison of opposing views and encouraged *"thinking from different perspectives"* (P31). By contrast, the Summarizing card was used sparingly, with some learners (e.g., P17) feeling that it was *"out of place"* during ongoing dialogue.
- **Knowledge proficiency determines exploration stance**. Topic familiarity shaped whether learners explored tentatively or argued with confidence. With limited knowledge, learners leaned on *Speculating, Reflecting,* and *Hypothesizing* cards to keep the dialogue moving and could *"offer a hypothesis or reflect on experience, even if I didn't know the exact answer"* (P21). With greater familiarity, learners strengthened responses through *Analogizing* and *Exemplifying* cards to clarify abstractions, as examples made answers *"more concrete"* (P28) and analogies *"more accessible"* (P1).
- **Individual preference influences effort intensity**. Personal style and preference also guided choices. Some participants (P2, P30, P31) favored concise, low-effort frames such as the bullet-point *Classifying* and *Organizing* cards to maintain fluency, which they described as improving hierarchy and helping to *"quickly sort the outline."* In contrast, several preferred more demanding strategies to build structured arguments, such as the *Targeting* card to make communicative intent explicit (P3, P21). Yet many others found these effortful strategies burdensome or misaligned with the dialogue flow, which explains the low uptake of cards such as Critiquing (P3, P5, P27, P28), Challenging (P21, P25), Countering (P2, P28, P31, P33), Clarifying (P19), and Implementing (P26).

4.4.2 Critical Thinking Scores. A paired-sample t-test revealed that the mean score of **critical thinking** in the experimental condition was significantly higher than in the control condition ($t_{33} = 6.675$, $p < .05$), with participants in the experimental condition showing stronger depth, breadth, and significance across the nine scoring criteria (see Figure 7).

Corroborated with the interviews, we derived three factors that influenced participants' critical thinking as follows.

- **Diverse Perspectives expands analytical breadth**. Learners indicated that the cards encouraged them to consider additional perspectives and extend their responses. As P10 noted, *"When I tried to use the Classifying card, I had to think of more ideas I could add, which expanded my answer."* Likewise, P23 remarked, *"Once I identified one advantage, the cards helped me generate further points to develop a more comprehensive answer"*.
- **Progressive argument increases reasoning depth**. Several participants reported moving from brief statements to more articulated, stage-by-stage reasoning. P25 explained, *"After seeing the Organizing card, I felt I could express the entire chain of thought in stages, from high tariffs to merchants, and then from merchants to consumers."* Similarly, P31 commented that the cards helped *"quickly clarify the outline of the answer" and present "more in-depth arguments,"* while P16 noted that the stepwise argument *"made it easier to build reasoning."*
- **Topical relevance improves significance focus**. The cards guided learners to stay on-topic and concentrate on essential points rather than drifting into redundant or tangential remarks. P2 stated that they *"clearly structure my thoughts, enabling me to avoid redundancy and highlight the key issue,"* while P22 reflected that *"I often lose track when answering long questions, but with the cards, I was able to keep my responses aligned with the main topic."*

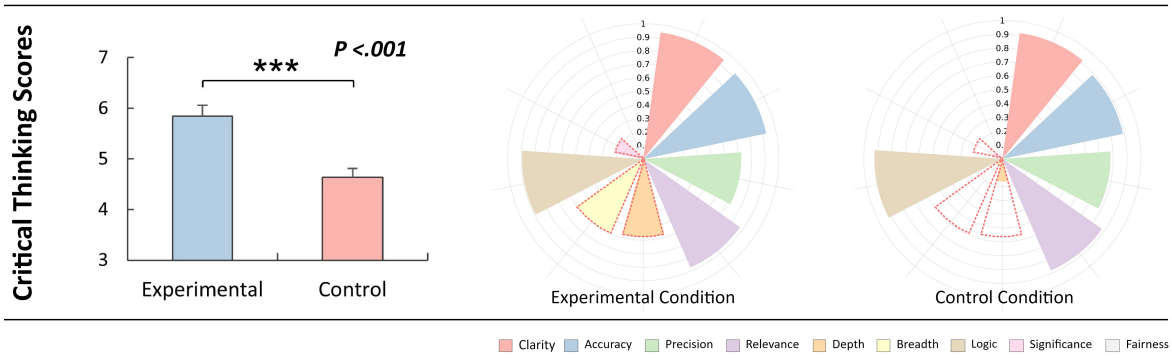


Figure 7: Results of the critical thinking scores between experimental and control conditions. The red dashed boxes highlight the rating criteria with marked differences. Error bars show standard error.

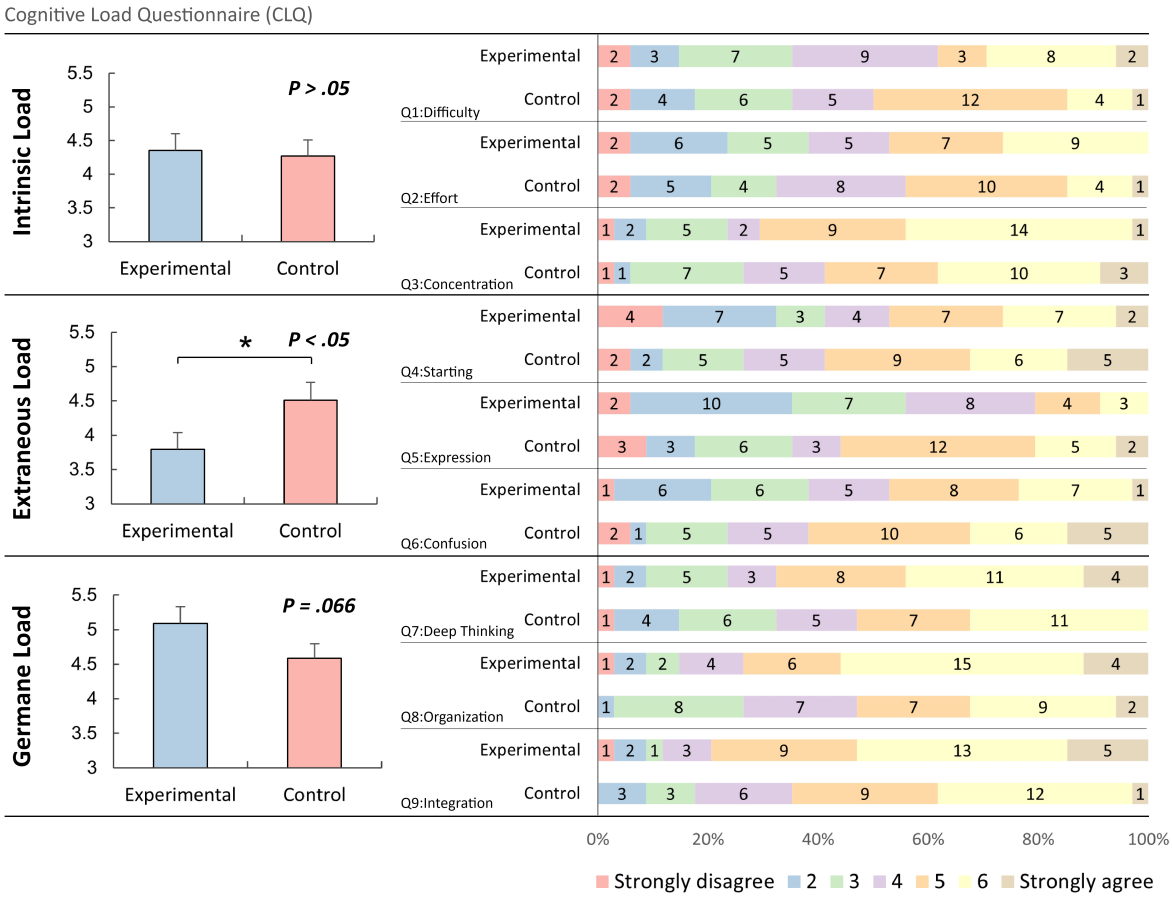


Figure 8: Results of the CLQ between experimental and control conditions. Error bars show standard error.

4.4.3 Cognitive Load Ratings. The comparative results for Intrinsic Load, Extraneous Load, and Germane Load across conditions are shown in Figure 8. H2a is supported: there was no significant difference in **Intrinsic Load** between the experimental and control conditions ($t_{33} = 0.348$, $p > .05$), since participants engaged with the same self-selected learning topics in both sessions. This suggests that the inherent complexity of the learning content remained stable across conditions.

Extraneous Load, defined as the unnecessary effort associated with organizing and articulating ideas, was significantly lower when learners had access to the Scaffolding Cards (t_{33} = -2.753, p

< .05). Thus, H2b is supported. Interview evidence attributed this reduction to two main factors:

- **Opening prompts facilitate response initiation.** The cards acted as initiating aids when participants struggled to begin their answers. P32 noted, *"When I don't know how to start, a card gives me a direction and saves time trying to think how to phrase it."* Similarly, P2 remarked that the scaffolds helped her *"think efficiently and organize language to give a try."* P19 recalled that when he had *"no clue how to begin,"* seeing a *Reflecting* card (*"My experience is..."*) suddenly gave him the inspiration to start.
- **Structure externalization directs content focus.** The cards helped learners concentrate more effectively on content rather than on the effort of phrasing or structuring their responses. For example, P22 stated, *"It helped me focus on the learning content,"* suggesting that having a ready-made linguistic structure reduced the mental load of composing responses from scratch. Likewise, P12 observed that the scaffolds reduced *"idle chatter"* and promoted a more purposeful, content-aligned response style.

Germane Load, which reflects the productive mental effort directed toward learning, showed a marginal increase in the experimental condition ($t_{33} = 1.904$, $p = .066$), indicating a trend but not statistical significance. Thus, H2c is not supported. Interview evidence attributed this tendency to two main factors:

- **Intentional thinking promotes active engagement**. The scaffolding strategies prompted intentional thinking rather than superficial replies. P21 noted that the cards sometimes *"forced or guided me to think"* more critically; P3 reported that frames such as *"I think... because..."* led her to probe underlying causes and motivations; P17 explained that *Questioning* and *Reflecting* cards prompted him to *"further think through the question and potential solutions"*.
- **Connection prompting deepens content understanding.** The cards cued learners to link ideas, causes, and examples, which consolidated and integrated content. P25 noted cards helped him *"unpack the reasoning step by step"*, and P7 described how this process *"opened up new knowledge"*. When topics were abstract, P6 reported that *"using an Analogizing card connecting ideas made my point more intuitive"*.

However, the lack of statistical significance of Germane Load may be explained by two limiting factors. First, **pre-formed ideas reduce scaffold utility**, since some learners already had clear ideas before answering. As P1 explained, *"if I already have an idea in my head, the cards are not that important,"* and P9 similarly noted, *"when facing the question, I basically came in with a complete answer most of the time."* Second, **habit misalignment undermines frame adoption**, making the sentence frames feel restrictive rather than supportive. P19 reflected that *"Many cards didn't fit the way I normally speak, and that probably influenced how I thought,"* while P16 emphasized a preference for a freer, divergent style, and P20 compared the cards to *"sentence-making drills in school essays,"* admitting he was not good at that format.

4.4.4 Learning Satisfaction Ratings. **Overall learning satisfaction** was significantly higher in the experimental condition ($t_{33} = 3.067$, $p < .05$), as shown in Figure 9, supporting H3 and indicating that participants were generally more engaged and satisfied with the learning dialogue when they had access to the Scaffolding Cards.

Interview evidence revealed three key factors that contributed to the positive ratings of the cards, as follows.

- **Dialogue ownership enhances learner autonomy**. The cards made the dialogue feel learner-driven rather than a passive interrogation. P9 described using the *Questioning* card to pose her own prompt, which *"stimulated my thinking"* and turned the exchange into a more dynamic conversation. P11 said that the *Reflecting* card helped him express *"my true thoughts,"* increasing the personal relevance and meaning of the interaction.
- **Preset logic improves expression clarity**. The cards reduced cognitive frustration and kept the conversation flowing by offering ready-to-use structures. P14 explained that the frames *"regulated my scattered thinking,"* enabling longer and more organized responses. P6 and P7 noted that the cards helped them avoid *"getting stuck"* or *"repeating themselves"*, sustaining the rhythm and interest of the exchange.
- **Comprehensive answers increase participation confidence**. Compared with fragmented, stop-start replies without cards, the frames supported completion of a coherent answer, making participants feel more informed and therefore more willing to contribute. For example, P14 reported *"greater expressive confidence"* once the frame was in place, making it easier to produce extended, well-organized responses.

Despite the higher overall satisfaction ratings, participants also noted several limitations, including absent card recommendations, cumbersome strategy navigation, insufficient content prompts, and limited progress awareness. By combining these concerns with participants' desired improvements, we coded and discussed four directions for design implications in section 5.2, including personalized scaffolding for context sensitivity, intuitive interaction for dialogue fluency, motivational mechanisms for sustained engagement, and learning analytics for timely reflection.

5 Discussion

A summary of our hypothesis testing results is presented in Table 4. For **RQ2** on how response strategies impact learning outcomes, the results of critical thinking showed significantly higher scores with the support of Scaffolding Cards (H1 supported). Interview evidence pointed to expanded analytical breadth, increased reasoning depth, and heightened significance expression as the main areas of improvement. In addition, usage patterns confirmed that learners' strategy choices were purposeful rather than random, shaped by dialogue context, knowledge proficiency, and individual preference. Taken together, **we concluded that structured response strategies improve the learning outcomes** during LLM-based Socratic instruction by contributing to improved critical thinking and personalized strategy use.

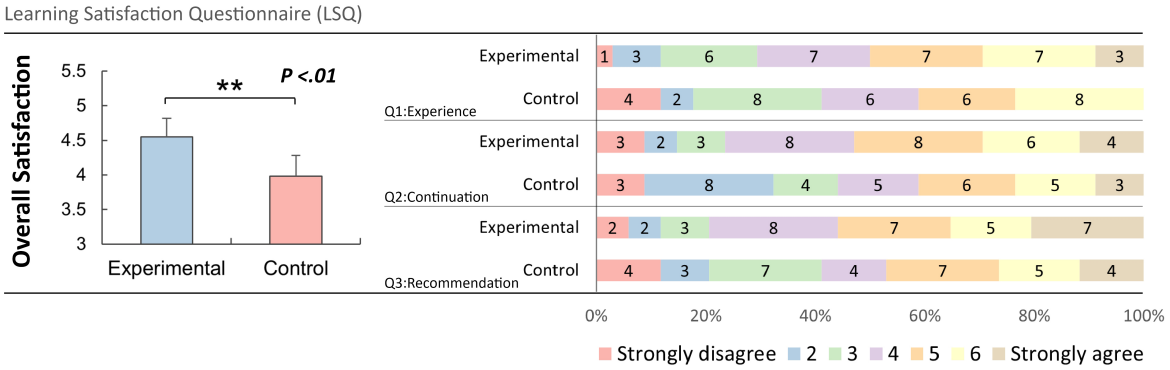


Figure 9: Results of the LSQ between experimental and control conditions. Error bars show standard error.

Table 4: Summary of the hypothesis outcomes.

Hypotheses	Descriptions	Support for Hypotheses
H1	Critical thinking scores will be **significantly higher** in the experimental condition than in the control condition	Supported
H2a	Intrinsic Load ratings will **not differ** between the experimental condition and the control condition.	Supported
H2b	Extraneous Load ratings will be **significantly lower** in the experimental condition than in the control condition.	Supported
H2c	Germane Load ratings will be **significantly higher** in the experimental condition than in the control condition.	Not Supported (marginal trend)
H3	Overall satisfaction ratings will be **significantly higher** in the experimental condition than in the control condition.	Supported

For **RQ3** on learners' perceptions of response strategies, results showed consistently favorable evaluations. Intrinsic Load did not differ across conditions (H2a supported), suggesting that the response strategies did not add extra difficulty. Extraneous Load was significantly reduced with cards (H2b supported), as interview evidence explained that learners benefited from response initiation support and content-focused attention, which shifted their effort from wording to meaning. Germane Load showed a marginal increase (H2c not supported). This tendency may reflect the cards' capacity to foster active engagement and deeper understanding, yet the effect was offset by learners' pre-formed ideas and by mismatches with personal expression habits, which limited consistent gains. Overall learning satisfaction was also higher with cards (H3 supported), with qualitative accounts pointing to an enhanced sense of autonomy, greater clarity of expression, and increased confidence during dialogue. Taken together, **we concluded that learners perceived the response strategies positively** according to the optimized cognitive load allocation and increased learning satisfaction.

5.1 Reconfiguring Scaffolding in LLM-based Socratic Instruction

This section discusses why Scaffolding Cards are effective in LLM-based Socratic instruction and what they contribute to scaffolding as a method. Building on the three design considerations in section 3.3.2, we examine their impact through three lenses: scaffolding with adaptation to learner states (DC1), scaffolding with preservation of learner agency (DC2), and scaffolding with transparent functions (DC3).

5.1.1 State-aware Scaffolding: Shifting from Tutor Prompts to Response States. The Scaffolding Cards shift the locus of scaffolding from tutor prompts to learner responses. Unlike successive human tutor prompts [65, 74] or LLM adaptive questioning [27, 38, 97], which depend on analyzing prior learner answers to generate follow-up questions, **the cards intervene within the same turn, supporting learners directly at their immediate response states.**

On the one hand, it reshapes usage patterns. Because scaffolding is now aligned with the response state, strategy selection naturally matched knowledge state: learners adopted exploratory strategies to sustain progress when uncertain and structure-oriented strategies to elaborate ideas when familiar, showing a competence-based calibration consistent with self-regulated learning [26]. On the other hand, it fosters critical thinking toward higher-order reasoning [32]. By acting at the point of response, the design broadened analysis by turning vague intuitions into multiple perspectives [44, 91], deepened reasoning by structuring multi-step explanations that reconstruct knowledge [90], and sharpened significance by steering learners to foreground essential points rather than fragmented replies [59].

As a result, this state-aware alignment realizes scaffolding for dialogue [44] in a response-centered way, enabling learners to move from tentative answers to reasoned arguments within a single dialogue turn without requiring multi-turn tutor prompting, thereby advancing a response-centered view of scaffolding in LLM-based Socratic instruction.

5.1.2 Agency-preserving Scaffolding: Shifting from Content Supply to Sentence Framing. Learner agency refers to students' capacity to act purposefully and constructively in their learning processes. To preserve it, Scaffolding Cards shift support from content supply to sentence framing, providing concise, optional, and editable frames. **This agency-preserving shift reduces formulation effort while leaving semantic choices to the learner, offering support without taking over the answer**.

At a behavioral level, learners tailored frame use to their style and task needs, choosing compact frames for fluency or more demanding frames to build multi-step arguments, which underscores the value of flexible options for different reasoning styles and motivational states [30]. At an experiential level, learners reported a stronger sense of autonomy because they could decide whether and how to use the frames, making the dialogue feel learner-driven. At a cognitive mechanism level, the frames optimized effort by reducing Extraneous Load through response initiation support and content-focused attention, while Germane Load was numerically higher, suggesting that freed capacity could be invested in constructive processing, consistent with cognitive offloading that reduces peripheral demands [3, 46] and frees working-memory resources [55] for understanding and transfer [80].

Taken together, agency-preserving scaffolding illustrates an alternative mode of support from what to say to how to construct a response in LLM-based Socratic instruction, which maintains learner initiative while still guiding the process of reasoning.

5.1.3 Function-transparent Scaffolding: Shifting from Implicit Cues to Explicit Messages. Unlike implicit cues, such as vague follow-up prompts, generic feedback, or subtle contextual signals that learners may miss [99], Scaffolding Cards make support explicit through specifying the function of a response strategy and how it can be applied. It transforms scaffolding into a metacognitive act [100] in LLM-based Socratic instruction, where **learners not only receive support but also become aware of the kind of reasoning they are enacting**.

Viewed from a design perspective, function transparency positions scaffolding itself as an interactional artifact, where support becomes visible and manipulable, enabling learners to appropriate the scaffolds as part of their own dialogic practice. This clarity reduces uncertainty and communicative friction while reinforcing a sense of competence, as learners effectively adapted strategies to situational demands such as question type and turn position and functioned the cards as flexible scaffolds [14] embedded in interaction rather than rigid templates. In addition, function transparency reshaped the affective tone of dialogue, shifting it from a stressful effort toward more engaged participation [40]. This shift was reflected in clearer articulation of ideas and greater confidence in contribution, aligning with fundamental psychological needs that sustain intrinsic motivation and engagement [20].

In sum, transparent functions transform scaffolding into explicit and manipulable guidance that makes reasoning visible, reduces uncertainty, and supports informed strategy use, thereby deepening learner engagement.

5.2 Design Implications for Inspiring Future LLM-based Socratic Platform

Based on participants' learning experience and desired improvements to response strategies and interaction methods, we propose design implications for embedding Scaffolding Cards into future LLM-based Socratic platforms. Figure 10 presents prototype illustrations of these implications, which aim to support more responsive and personalized learner-LLM interaction. Four main areas of improvement are highlighted:

Personalized scaffolding for context sensitivity. In our study, participants selected clusters and strategies themselves, but found manually browsing all cards to be effortful and expressed an expectation for personalized recommendations. Effective scaffolds should respond dynamically to task structure and evolving learner needs [72], allowing users to select or customize cards according to their own preferences, ensuring that the response process remains centered on the learners' own thinking while preserving the flow of Socratic dialogue. Participants echoed this need: P4 suggested that *"the system could recommend the most suitable cards based on the question,"* and P12 hoped for *"automatic suggestions"* based on current ideas or past responses. A practical implication is to provide a manageable set of cards with options for ranking, expanding, or customizing, enabling personalization over time while retaining learner agency and keeping learners actively engaged in directing their own inquiry.

Intuitive interaction for dialogue fluency. Interfaces should minimize choice overload and peripheral effort so that scaffolding remains a support rather than a distraction. Showing a small, diverse subset of strategies at a time reduces cognitive burden and helps maintain dialogue flow [34]. Learners emphasized visual layout and accessibility (P3, P13, P15, P16) and suggested one-click insertion for smoother use (P5, P12), more compact displays (P8), and functional grouping of cards (P2) to streamline interaction and reduce interface overhead. To further lower friction, provide quick ways to refresh or switch to a new batch of strategies and to view illustrative examples, keeping attention on content rather than interface work. In practice, surface a short-ranked list with one-click insertion, an expand-to-view-all option, and a refresh control for a new batch.

Motivational mechanisms for sustained engagement. Scaffolding should also address motivation. Learners often avoid complex strategies because they anticipate high effort rather than a lack of ability. Interviews indicated a need for content support that lowers this threshold: clearer usage examples (P3, P13) and conceptual prompts or knowledge cues when topics were unfamiliar, which sustained engagement and persistence (P4, P20, P24). To complement this, interfaces can incorporate lightweight, game-inspired incentives [81] that remain consistent with the Socratic focus on reflective reasoning. For example, award points for selecting increasingly challenging frames and allow points to be redeemed for limited, optional support such as hints or partial answers. These

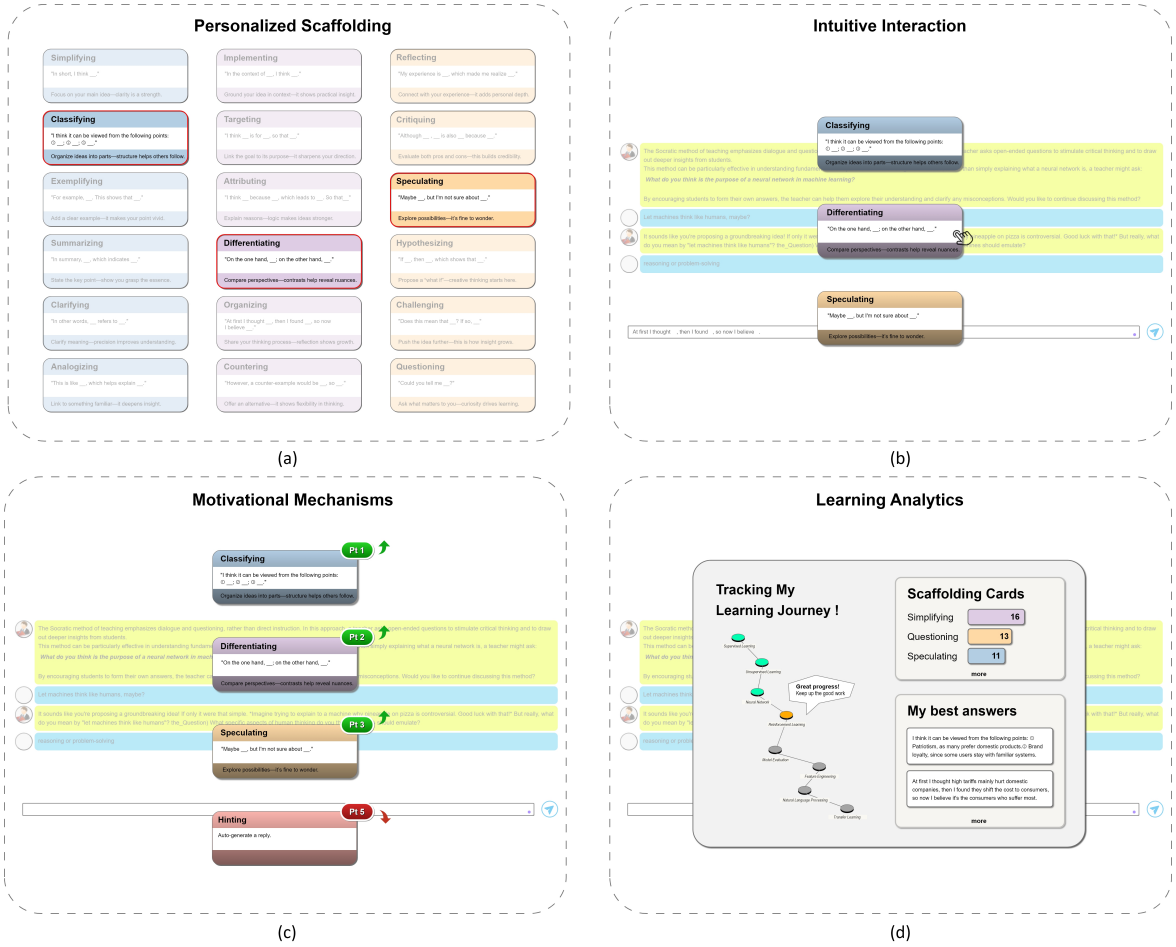


Figure 10: Illustrative prototypes of four design directions for Scaffolding Card integration: a) Personalized scaffolding, b) Intuitive interaction, c) Motivational mechanisms, and d) Learning analytics.

mechanisms are to maintain participation, nudge deeper reasoning, and channel effort toward learning.

Learning analytics for timely reflection. Open-ended dialogue can feel endless when progress is not visible, underscoring the value of embedding lightweight analytics in scaffolding systems [91]. Interfaces that summarize progress, display the frequency of strategy use, and highlight effective examples can support goal setting and self-monitoring, which are central to self-regulated learning [48]. By making strategy choices and their outcomes transparent, learners can reflect on how their interaction patterns shape the dialogue and adjust scaffold use accordingly. For example, P5 wanted to track which cards were used most and whether they improved responses, and P6 proposed post-dialogue prompts such as *"next time you could try this card"* to support ongoing skill development. From a system perspective, analytics can detect underused or overused strategies and recommend balanced adjustments, helping to scaffold remaining responsive over time.

5.3 Reclaiming Learner Agency from LLM-driven Education

LLM-based Socratic instruction is valuable not for supplying answers, but for shifting the responsibility for thinking back to the learner [38, 92, 97]. Our stance is response-centered: we advance LLM-based Socratic instruction by treating the system not merely as a conversational partner but as a scaffold for learners' responses, supporting how ideas are constructed while leaving meaning with the learner. In doing so, scaffolding serves as a design mechanism for "taking the agency back" from LLM-led tutoring to human-led learning, a shift in locus of cognition where support targets the organization, justification, and extension of the learner's own ideas. Seen in this light, our Scaffolding Cards operationalize a principled balance, **making the LLM present as a catalyst for reasoning progression, while keeping the human responsible for claims, warrants, and choices**.

Cognitive-level governance of agency. In our account, cognitive level coverage is not merely breadth but a governance mechanism for learner agency. By binding each scaffold to a specific cognitive act in Bloom's revised taxonomy [5] and invoking it at the

learner's response state, the system allocates accountability for the psychological work to the human rather than to the LLMs. Because scaffolds are selectable at the moment, dynamic leveling within a single turn enables learners to escalate (e.g., from Analyze to Evaluate) or de-escalate (e.g., back to Understand) as needed. This micro-selection is itself an exercise of agency, anchoring higher-order work in visible, learner-performed steps within a mixed-initiative interaction [37]. The frames also render the level metacognitively visible of "which level am I working at, and why here?", thereby making level transitions an object of learner monitoring and control [100]. In short, **our response strategies provide a portable pathway for mastering all six cognitive levels in future LLM-driven education, keeping the learner in charge of destination and ascent while mitigating the risk of plateauing at lower-order cognition under heavy reliance on LLM-delivered answers.**

Discipline-agnostic transfer of agency. As the strategies encode reasoning processes rather than domain knowledge, the same repertoire generalizes across disciplines and task settings. In our study, participants selected topics spanning technical and social domains, yet the strategies remained effective because they target how to think rather than what to conclude. This process-first design yields a scaffold usable across disciplines that helps learners structure responses in unfamiliar territories, supports transfer of strategy use, and offers instructors a common language for guiding higher-order work across courses. At the same time, it remains adaptable: frames can be selected or combined to respect domain conventions and rhetorical norms, preserving expressive diversity rather than standardizing voices. In sum, **future LLM-driven education would benefit from incorporating a scaffold that generalizes across disciplines, sustaining agency precisely when knowledge and conventions vary, rather than fully relying on direct answers from LLMs**.

5.4 Limitations and Future Work

While this study provides promising evidence for the effectiveness of Scaffolding Cards in LLM-based Socratic instruction, several limitations remain and should be addressed in future research. 1) The current system provided a fixed set of cards to all users without personalized recommendations. While the manual approach allowed for controlled comparisons, it did not reflect the dynamic interactions between various learners and LLMs. Future research should explore the effectiveness of integrating Scaffolding Cards into systems that support learner-driven personalization, such as allowing users to select or customize cards according to their preferences, thereby maintaining a learner-centered process. 2) Effective scaffolding design should enable gradual fading due to the potential risk of weakening the learner's agency in long-term usage. Future work should examine the use of Scaffolding Cards across multiple sessions and diverse learning contexts to assess potential overreliance on cards, adaptability to various instructional goals, and long-term effects on learners' reasoning skills and cognitive development. 3) The familiarity was controlled by a within-subject design in this study. Therefore, its impact on LLM-based Socratic instruction remains unclear and warrants further study. Investigating how scaffolding supports learners at different stages, such as pre-class preparation with low-familiarity and post-class review with high-familiarity, would help determine when and for whom it is most effective. By addressing these limitations, future work can advance the design and deployment of scaffolding approaches in LLM-based education, supporting more diverse, complex, and sustainable learner-LLM interaction.

6 Conclusion

This study addressed the challenge of learners producing superficial responses in LLM-based Socratic instruction by developing and evaluating Scaffolding Cards on the SPL system. We first derived 18 Scaffolding Cards from a formative study, organizing sentence-framing strategies into three clusters that support idea organization, argument construction, and reflective thinking (RQ1). A subsequent mixed-methods evaluation showed that the cards improved critical thinking, optimized cognitive load allocation, and increased learning satisfaction relative to a no-scaffold control, while enabling personalized strategy use during dialogue (RQ2, RQ3). Leveraging Scaffolding Cards as both intervention and research probes, we contributed reusable research assets that future studies can build upon. Our findings reconfigure scaffolding for LLM-based Socratic instruction into state-aware, agency-preserving, and function-transparent support, offering actionable implications for designing responsive and personalized scaffolding to strengthen learner-LLM interaction.

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A Appendix: Interview Script

A.1 Interview Protocol of The Formative Study

The interview questions were as follows:

1. Why did you choose this learning topic, and how familiar are you with it?
2. What were the main challenges you encountered while responding to questions in the dialogue? Can you provide specific examples of moments when you felt stuck or uncertain, and were there also times when the response felt smooth or confident by contrast?
3. If tools or methods were available to assist you in responding to questions, how do you think they would impact your response process or outcomes? What specific features or functionalities would you expect from such support tools?

A.2 Interview Protocol of The User Study

The interview questions were as follows:

1. Which cards did you use most often, and which did you rarely use or avoid? Please list examples in both groups and explain why.
2. Did the cards help you construct stronger answers? How did you use them to develop structuring, reasoning, and ideating? Please give examples where you revised or deepened the answer.
3. Compared with answering without cards, how did the cards change your effort to formulate and organize language and your state of thinking about the question? Please give dialogue examples.
4. Overall, how did the cards affect your learning satisfaction during the dialogue? What felt helpful or frustrating?
5. What improvements would you like for the response strategies or the interaction method to better support you?

B Appendix: Order Effects

Independent samples t-tests were conducted to compare the two order groups and assess the effectiveness of counterbalance. Order 1 consisted of participants completing the control condition first, while Order 2 consisted of participants completing the experimental condition first. As shown in Table 5, no significant differences were found across any dependent variables ($p > .05$), confirming that counterbalance effectively mitigated potential order effects.

C Appendix: Time Difference

Linear regressions were conducted using the time difference between the experimental and control conditions as the predictor and the difference scores of each dependent variable as the criterion. As shown in Table 6, none of the regressions were significant ($p > .05$), indicating that the additional time spent using the cards did not systematically influence the dependent variables.

D Appendix: Critical Thinking Rubric

This critical thinking rubric consists of nine evaluation criteria, each scored on a binary scale (0 = skill not demonstrated; 1 = skill demonstrated), as shown in Table 7.

E Appendix: Questionnaire

This appendix lists the complete items used in the questionnaires. All items were rated on a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree).

E.1 Cognitive Load Questionnaire (CLQ)

The items used to measure Intrinsic Load are as follows:

1. The questions in this phase presented a certain level of learning difficulty for me.
2. Understanding the questions in this phase required considerable mental effort.
3. Completing the tasks in this phase required a high level of concentration.

The items used to measure Extraneous Load are as follows:

1. When answering the questions in this phase, I often didn't know where to start.
2. It was somewhat difficult for me to express my ideas during this phase.
3. I felt confused or mentally disorganized while answering the questions.

The items used to measure Germane Load are as follows:

1. Answering the questions in this phase prompted me to think deeply about the topic.
2. Throughout this process, I was able to clearly organize and express my ideas.
3. During the answering process, I was motivated to integrate and articulate my understanding.

E.2 Learning Satisfaction Questionnaire (LSQ)

The items used to measure learning satisfaction are as follows:

1. Overall, I am satisfied with the learning experience in this phase.
2. I would like to continue using this type of learning approach in the future.
3. I would recommend this type of learning experience to others if available.

Table 5: The effectiveness of counterbalance across dependent variables.

Variables	Conditions	Order 1 Mean (SD)	Order 2 Mean (SD)	t	df	p
Critical Thinking	Experimental	5.6 (1.5)	6.0 (1.0)	-.933	32	.358
	Control	4.3 (1.3)	4.9 (0.3)	-1.861	17.883	.079
Intrinsic Load	Experimental	4.3 (1.6)	4.5 (1.3)	-.393	32	.697
	Control	4.1 (1.3)	4.5 (1.5)	-.834	32	.410
Extraneous Load	Experimental	3.7 (1.5)	3.9 (1.4)	-.357	32	.724
	Control	4.3 (1.6)	4.7 (1.4)	-.904	32	.373
Germane Load	Experimental	4.9 (1.5)	5.3 (1.3)	-.764	32	.450
	Control	4.6 (1.3)	4.5 (1.2)	.185	32	.854
Learning Satisfaction	Experimental	4.5 (1.6)	4.6 (1.6)	-.357	32	.723
	Control	4.0 (1.7)	4.0 (1.9)	.064	32	.949

Table 6: Regression results for the effects of time difference on dependent variables.

Dependent Variable	B	SE B	Beta	t	p
Critical Thinking	-.049	.053	-.160	-.918	.365
Intrinsic Load	-.020	.067	-.053	-.301	.765
Extraneous Load	-.049	.077	-.112	-.640	.527
Germane Load	.081	.077	.182	1.047	.303
Learning Satisfaction	.009	.055	.029	.165	.870

Table 7: Critical thinking evaluation rubric.

Criteria	Score: 0	Score: 1
Clarity	The response is out of scope and incomprehensible.	The response is highly relevant and comprehensible.
Accuracy	The response does not align with existing scientific theories.	The response aligns with established scientific theories.
Precision	The response uses vague or non-academic language.	The response uses precise and academic language.
Relevance	The response is irrelevant to the question.	The response is highly relevant to the question.
Depth	The response includes only superficial information.	The response presents insights or deeper reasoning.
Breadth	The response offers a single perspective.	The response considers multiple perspectives.
Logic	The response is incoherent or internally inconsistent.	The response is logically structured and coherent.
Significance	The response lacks critical or important points.	The response addresses key or significant issues.
Fairness	The response is biased or overly subjective.	The response is balanced and objective.

F Appendix: Interview Analysis

Table 8 shows the summary of interview analysis findings of the user study.

Table 8: Summary of interview analysis findings.

Dimensions	Codes	Participants
Usage patterns	Dialogue context calibrates immediate purpose	P7, P17, P19, P21, P27, P31, P34
	Knowledge proficiency determines exploration stance	P1, P21, P28
	Individual preference influences effort intensity	P2, P3, P5, P19, P21, P25, P26, P27, P28, P30, P31, P33
Critical thinking	Diverse Perspectives expands analytical breadth	P10, P23
	Progressive argument increases reasoning depth	P16, P25, P31
	Topical relevance improves significance focus	P2, P22
Cognitive load	Opening prompts facilitate response initiation	P2, P19, P32
	Structure externalization directs content focus	P12, P22
	Intentional thinking promotes active engagement	P3, P17, P21
	Connection prompting deepens content understanding	P6, P7, P25
	Pre-formed ideas reduce scaffold utility	P1, P9
	Habit misalignment undermines frame adoption	P16, P19, P20
Learning satisfaction	Dialogue ownership enhances learner autonomy	P9, P11
	Preset logic improves expression clarity	P6, P7, P14
	Comprehensive answers increase participation confidence	P14
Desired improvements	Personalized scaffolding for context sensitivity	P4, P12
	Intuitive interaction for dialogue fluency	P2, P3, P5, P8, P12, P13, P15, P16
	Motivational mechanism for sustained engagement	P3, P4, P13, P20, P24
	Learning analytics for timely reflection	P5, P6